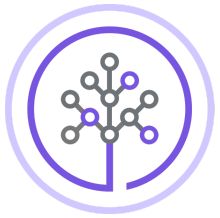


Lab: Systems Architect Skills Matrix from Job Listings



Skills
Network

Estimated time needed: 20 minutes

Overview

In this lab, you will analyze real-world **Systems Architect** job listings to identify, categorize, and evaluate the key skills and competencies required in the industry. You will create a comprehensive skills matrix that maps technical and soft skills across different industry sectors and experience levels.

This lab helps you demonstrate your ability to research market demands, analyze job requirements, and create structured documentation for career planning and skill development.

Learning objectives

- Analyze and extract key skills from systems architect job postings
 - Categorize technical and soft skills into meaningful groups
 - Identify industry-specific requirements and common skill patterns
 - Create a structured skills matrix for career planning
 - Evaluate personal skill gaps and development opportunities
-

Step 1: Download resources and setup

Download required materials

Skills matrix Excel template:

-  [Right-click here](#) and 'Open link in new tab' to download the skills matrix Excel template.

Template includes:

- Skills matrix worksheet with pre-defined categories
- Job analysis worksheet for detailed breakdown
- Personal assessment worksheet
- Career planning worksheet
- Data validation and formulas for analysis

Setup instructions:

- Download the Excel template above
 - Open the template in Excel
 - Keep the job listings (Step 2) open in a separate window for reference
-

Step 2: Analyze real-world job listings

Review the following five **Systems Architect** job postings sourced from major job platforms. Pay attention to both technical requirements and soft skills mentioned.

Job listing A: Senior Cloud Systems Architect - Financial Technology

Company: JPMorgan Chase & Co.

Location: New York, NY | Remote Hybrid

Salary: \$165,000 - \$225,000

Experience: 8+ years

Key responsibilities:

- Design and implement large-scale distributed systems handling 10M+ daily transactions
- Lead architecture reviews and technical decision-making for cloud migration initiatives
- Collaborate with product teams to translate business requirements into scalable technical solutions

Required skills:

- Expert-level AWS services (EC2, S3, RDS, Lambda, EKS, VPC)
- Programming proficiency in Java 11+, Python 3.x, or Scala
- Deep understanding of microservices architecture and domain-driven design
- Experience with containerization (Docker, Kubernetes, Helm charts)
- Message queuing systems (Apache Kafka, RabbitMQ, Amazon SQS)
- Database design: PostgreSQL, MongoDB, Redis, Apache Cassandra
- API design (REST, GraphQL, gRPC) and gateway patterns
- Infrastructure as Code (Terraform, CloudFormation, Ansible)
- CI/CD pipelines (Jenkins, GitLab CI, AWS CodePipeline)
- Monitoring and observability (Prometheus, Grafana, ELK stack, AWS CloudWatch)
- Security frameworks and compliance (SOX, PCI-DSS, OAuth 2.0, SAML)

Preferred qualifications:

- AWS Solutions Architect Professional certification
 - Experience with high-frequency trading systems
 - Knowledge of financial regulations and risk management
 - Strong communication skills for executive-level presentations
 - Agile/Scrum Master certification
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Job listing B: Enterprise Systems Architect - Healthcare Technology

Company: Epic Systems Corporation

Location: Madison, WI

Salary: \$145,000 - \$190,000

Experience: 10+ years

Key responsibilities:

- Architect healthcare interoperability solutions supporting 200+ hospital clients
- Design HIPAA-compliant systems for patient data management and clinical workflows
- Lead integration projects with legacy healthcare systems and modern cloud platforms

Required skills:

- Healthcare standards: HL7 FHIR, CDA, DICOM, IHE profiles
- Enterprise architecture frameworks (TOGAF 9.2, Zachman, SABSA)
- Integration patterns: ESB, API Management, ETL/ELT processes
- Microsoft Azure cloud services (Azure Health Data Services, FHIR Server)
- .NET Framework/.NET Core, C#, PowerShell scripting
- SQL Server, Oracle databases, data warehousing concepts
- HIPAA, HITECH compliance and healthcare cybersecurity
- Clinical workflow analysis and business process modeling
- Stakeholder management across clinical and technical teams
- Technical documentation and architecture diagramming (Visio, Lucidchart)

Preferred qualifications:

- Healthcare IT certifications (CPHIMS, HIMSS)
 - TOGAF 9 certification
 - Experience with Epic, Cerner, or Allscripts implementations
 - Clinical background or healthcare operations knowledge
 - PMP or equivalent project management certification
-

Job listing C: Lead Systems Architect - E-commerce Platform

Company: Shopify Inc.

Location: Toronto, ON | San Francisco, CA

Salary: \$180,000 - \$250,000 CAD

Experience: 7+ years

Key responsibilities:

- Scale architecture supporting 1M+ merchants and billions of annual transactions
- Design real-time inventory and payment processing systems
- Lead technical strategy for global expansion and multi-tenant architecture

Required skills:

- Ruby on Rails, Go, or Node.js for high-performance web applications
- Event-driven architecture and asynchronous processing
- Google Cloud Platform (GKE, Cloud SQL, Pub/Sub, BigQuery)
- Database sharding and partitioning strategies
- Content Delivery Networks (CloudFlare, Fastly)
- Payment gateway integrations (Stripe, PayPal, Square)
- Real-time analytics and data streaming (Apache Kafka, Google Dataflow)
- Load balancing and auto-scaling technologies
- A/B testing frameworks and feature flag management
- Mobile API design for iOS and Android applications
- Performance optimization and capacity planning
- Site reliability engineering practices

Preferred qualifications:

- Experience with merchant-facing platform development
 - Google Cloud Professional Architect certification
 - Knowledge of international e-commerce regulations
 - Open-source contribution history
 - Startup or high-growth company experience
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Job listing D: IoT Systems Architect - Manufacturing Technology

Company: Siemens Digital Industries

Location: Detroit, MI | Cincinnati, OH

Salary: \$155,000 - \$205,000

Experience: 8+ years

Key responsibilities:

- Design Industrial IoT architectures for smart manufacturing environments
- Integrate operational technology (OT) with information technology (IT) systems
- Develop edge computing solutions for real-time manufacturing analytics

Required skills:

- Industrial protocols: OPC-UA, Modbus, MQTT, CoAP
- Edge computing platforms (Azure IoT Edge, AWS IoT Greengrass)
- Time-series databases (InfluxDB, TimescaleDB, Historian systems)
- Manufacturing Execution Systems (MES) and Enterprise Resource Planning (ERP)
- Cybersecurity for industrial environments (IEC 62443, NIST frameworks)
- Programming: C++, Python, ladder logic (PLC programming)
- Network architecture for industrial environments (TSN, deterministic Ethernet)
- Machine learning for predictive maintenance and quality control
- SCADA systems and human-machine interfaces (HMI)
- Lean manufacturing principles and Industry 4.0 concepts
- Project management for cross-functional engineering teams

Preferred qualifications:

- Professional Engineer (PE) license
 - Certified in industrial cybersecurity (CISSP, GICSP)
 - Experience with Siemens, Rockwell, or Schneider Electric platforms
 - Six Sigma certification
 - German language skills (advantage for global teams)
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Job listing E: DevOps Systems Architect - Gaming Industry

Company: Unity Technologies

Location: Austin, TX | Remote

Salary: \$170,000 - \$230,000

Experience: 6+ years

Key responsibilities:

- Architect build and deployment systems for game development workflows
- Design scalable backend systems for multiplayer game services
- Optimize CI/CD pipelines for mobile and console game releases

Required skills:

- Container orchestration (Kubernetes, Docker Swarm, ECS)
- Multi-cloud architecture (AWS, GCP, Azure hybrid strategies)
- Game development tools and Unity3D engine architecture
- Real-time multiplayer networking and game server management
- CDN optimization for game content delivery
- Performance profiling and optimization for gaming workloads
- Monitoring for gaming metrics (latency, framerate, user engagement)
- Mobile deployment pipelines (iOS App Store, Google Play, console platforms)
- Version control at scale (Git LFS, Perforce for large assets)
- Analytics and telemetry systems for game metrics
- DevSecOps practices and vulnerability management

Preferred qualifications:

- Game development background or computer graphics knowledge
- Experience with AAA game release cycles
- Kubernetes certification (CKA, CKAD)
- Understanding of game monetization and live-ops

Analysis instructions:

Read each job listing carefully and note the variety of technical skills, industry-specific knowledge, and soft skills required across different sectors.

Step 3: Extract and categorize skills

Skill extraction process

Using your downloaded Excel template, systematically extract skills from each job listing and categorize them according to the following framework:

Primary categories:

1. **Cloud platforms & services**
2. **Programming languages & frameworks**
3. **Architecture patterns & design**
4. **Database technologies**
5. **DevOps & infrastructure**
6. **Industry-specific knowledge**
7. **Security & compliance**
8. **Communication & leadership**
9. **Certifications & credentials**
10. **Tools & technologies**

Instructions:

- Mark an 'X' in the corresponding job column for each skill mentioned
- Count frequency across all five job listings
- Note any industry-specific variations in the comments section

Skills matrix template structure:

Skill Category	Specific Skills	Job A	Job B	Job C	Job D	Job E	Total Frequency	Priority Level	Personal Rating (1-5)	Development Need
Cloud Platforms	AWS						0			
Cloud Platforms	Azure						0			
Cloud Platforms	Google Cloud Platform						0			
Programming Languages	Java						0			
Programming Languages	Python						0			
Programming Languages	C#						0			

Step 4: Frequency analysis and pattern recognition

Identify market trends

Complete the frequency analysis in your template:

Analysis tasks:

1. **Calculate skill frequency:** How many jobs mention each skill?
2. **Identify universal skills:** Which skills appear in 4+ job listings?
3. **Find industry-specific skills:** Which skills are unique to certain sectors?
4. **Rank by priority:** Assign priority levels (High/Medium/Low) based on frequency and market demand

Pattern recognition questions:

- Which technical skills are most consistently required?
- How do soft skill requirements vary by industry?
- What certifications provide the most career value?
- Which programming languages are in highest demand?

Analysis framework:

Universal skills (High priority):

- Skills appearing in 4-5 job listings
- Core competencies for all Systems Architects

Common skills (Medium priority):

- Skills appearing in 2-3 job listings
- Important for specific career paths

Specialized skills (Low-medium priority):

- Skills appearing in 1 job listing
 - Industry-specific or emerging technologies
-

Step 5: Personal skills assessment

Self-evaluation exercise

Using the "Personal Rating" column in your template:

Assessment tasks:

Rate your current skills (1-5 Scale):

- **1** = No experience
- **2** = Basic knowledge/exposure
- **3** = Working knowledge
- **4** = Proficient/experienced
- **5** = Expert/can teach others

Identify development priorities:

For each skill category, determine:

- **Immediate needs** (Skills required for current job market)
- **Growth opportunities** (Skills for career advancement)
- **Nice-to-have** (Skills that differentiate you)

Create your learning plan:

1. List 3 high-priority skills to develop in the next 6 months
2. Identify 2 certifications that align with your career goals
3. Choose 1 industry domain to specialize in

Reflection questions:

- Where are your strongest skill areas?
- What gaps exist between market demands and your current abilities?
- Which skills would provide the highest ROI for your career?
- How do your goals align with industry trends?

Personal assessment template:

Current skill inventory:

Technical skills:

- Cloud platforms: ____/5
- Programming: ____/5
- Architecture design: ____/5
- Database technologies: ____/5
- DevOps/infrastructure: ____/5

Soft skills:

- Communication: ____/5
- Leadership: ____/5
- Project management: ____/5
- Business analysis: ____/5

Gap analysis:

High-impact skills to develop:

1. _____
2. _____
3. _____

Certification Goals:

1. _____
2. _____

Conclusion

Congratulations! You have completed the **Systems Architect Skills Matrix Analysis** lab.

In this lab assignment, you successfully completed the following tasks:

- ☒ Analyzed five real-world Systems Architect job listings across different industries
- ☒ Extracted and categorized over 50 technical and soft skills
- ☒ Identified market trends and skill frequency patterns
- ☒ Conducted a personal skills assessment and gap analysis
- ☒ Developed critical thinking skills for career planning in the technology field

Key takeaways:

The Systems Architect role varies significantly across industries, but core competencies in cloud platforms, programming, and communication remain consistent. Your personal development should balance universal skills with industry-specific expertise to maximize career opportunities.

Next steps:

Use your completed skills matrix as a living document to track your professional development and guide your learning investments throughout your career.

General tips

1. Analysis challenges:

- Focus on explicit skills mentioned rather than implied competencies
- When in doubt about categorization, choose the most relevant category
- Don't overthink the frequency analysis - clear patterns will emerge

2. Personal assessment:

- Be honest about your current skill levels
- Consider both formal training and practical experience
- Ask colleagues or mentors for input on your skill assessments

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